already infinite can’t get any bigger. What did increase are the separations between objects like galaxies (once they formed), as you can see by looking from left to right in Figure 8.11b. An observer like you or me, looking out from one galaxy or another, would see surrounding galaxies all rushing away, just as Hubble discovered.

Bear in mind that this example of infinite flat space is far more than academic. We will see that there is mounting evidence that the overall shape of space is not curved, and since there is no evidence as yet that space has a video game shape, the flat, infinitely large spatial shape is the front-running contender for the large-scale structure of spacetime.

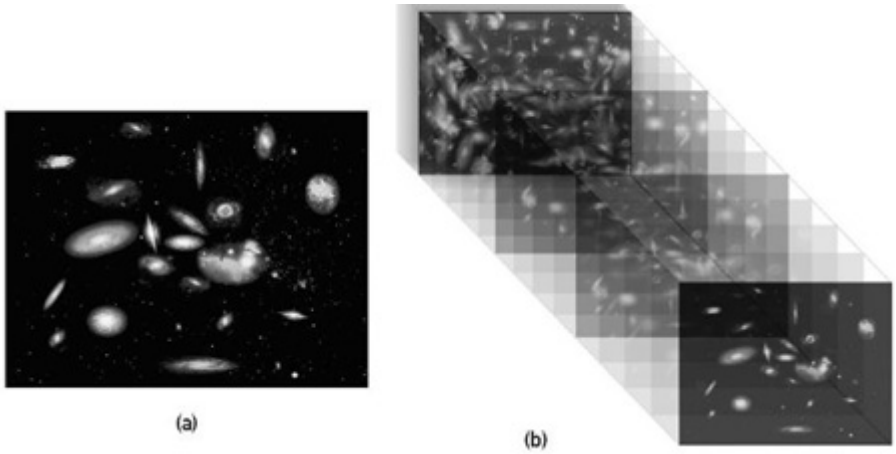


Figure 8.11 **(**a**)** Schematic depiction of infinite space, populated by galaxies. **(**b**)** Space shrinks at ever earlier times—so galaxies are closer and more densely packed at earlier times—but the overall size of infinite space stays infinite. Our ignorance of what happens at the earliest times is again denoted by a fuzzy patch, but here the patch extends through the infinite spatial expanse.

Cosmology and Symmetry

Considerations of symmetry have clearly been indispensable in the development of modern cosmological theory. The meaning of time, its applicability to the universe as a whole, the overall shape of space, and even the underlying framework of general relativity, all rest on foundations of symmetry. Even so, there is yet another way in which ideas of symmetry have informed the evolving cosmos. Through the